

STM32F103CBT6

STM32F103 custom development board

This project contains custom development board based on the STM32F103 microcontroller. This board created using kicad 9.0 software with proper routing, signal integrity, differential pair routing of USB connection.

Features

STM32F103 microcontroller

Onboard 3.3V linear voltage regulator(AMS1117-3.3)

Onboard Led indication on 3.3V power supply

Micro USB interface for power and communication

SWD interface for debugging and programming

UART and I2C breakout headers

Decoupling capacitors placed for stable operation

External crystal oscillator with load capacitors

MCU reset switch

Clean PCB layout with proper grounding and differential routing for micro USB connection

Board size: 35.5 mm X 23.6 mm

LDO power supply section

Pinout of LDO AMS1117-3.3

Vin

Vout

Gnd

Both vin and vout need decoupling capacitors connected with it

Gnd connects directly with ground

Input voltage= 4.5V to 12V DC

Output voltage=3.3V

Here LDO is linear voltage regulator it converts the extra voltage into heat
So as the heat is generating the efficiency decreases

Here 5v to 3.3v LDO conversion

Efficiency = v_{out}/v_{in}

= $3.3/5$

=66%

AMS1117-3.3

This LDO will give output as 3.3 v

As its voltage drop is high $\geq 1V$

Here both V_{in} and V_{out} pins of the LDO needed decoupling capacitors

In datasheet 10 μ F decoupling capacitors mentioned which is minimum value of capacitor need to use

So for good stability 22 μ F is used

This generated 3.3V output of LDO is goes into MCU power on pin

By giving 3.3V the MCU gets power on

There is one LED with 1.5k Ω resistor protection is connected at 3.3v output of an LDO to identify that 3.3v is available in LDO

USB_B micro

Pinout of USB B micro connector

D+ for data +

D- for data -

GND for ground

Vbus for +5v from USB

Shield= connects with gnd

Shield is metal holo part of the USB

Or shield can be connected with one resistor and one capacitor value for good ESD protection

ID for different mode

ID=floating= device mode

ID=connect with GND= host mode

In device mode the host will control to USB

Usb will respond to the host (PC)

Host mode

In this host will control everything

Pin 32 and pin33 of stm32f103 is used for USB connection

WHY PULL UP RESISTOR ON USB_D+ LINE REQUIRED

This pull up resistor tells to the host that device is connected

Without pull up resistor the host (PC) won't know that device is connected

If pull up resistor is connected with D+ and D-

If pull up resistor is connected with D+ then USB will work in full speed (12 mbps)

If pull up resistor is connected with D- then USB will work in low speed device (1.5 mbps)

STM32F103 MCU

Pinout

vdd= digital power 3.3v

vss=gnd

vdda= analog power 3.3v

vssa= gnd

NRST= reset (active low)

BOOT0= boot mode select

PB2 (BOOT1) = secondary boot config

Clock pins

PD0= oscillator input= crystal input

PD1= oscillator output= crystal output

Connect 4 to 16mhz external crystal range

Port A(PA0 to PA15)

ADC input, USART, SPI etc

Port B(PB0 to PB15)

I2c, spi, gpio

Port C(PC13 to PC15)

Low power pins , used for LED connection

Switch connection

PA9= USART TX

PA10= USART RX

PB6= I2C SCL

PB7=I2C SDA

PA5= SPI SCK

PA6= SPI MISO

PA7= SPI MOSI

PA13= SWDIO

PA14=SWCLK

Used for programming via st link

PC14

PC15 CRYSTAL

VBAT= back up battery connection

If battery is not connected then connect vbat with your input voltage either vdd or VDDA

Name structure of STM32

STM32F103

This 1 indicate low power speed

STM32F401

This 4 indicate higher speed compare to STM32F103

STM32L= L for ultra low power

STM32WB= WB indicates for wireless connection supported MCU

VDD pin of the MCU

Each vdd pin required decoupling capacitors connected with it

0.1uf for each vdd pin to remove high frequency noise

10uf for bulk energy storage and to remove low frequency noise

VDDA pin for analog input

VDD for digital power

Why VDD and VDDA two separate supply given in MCU

If VDD and VDDA is not separate out then digital noisy signal enter into analog circuits which cause wrong measurement of the ADC value

This occurs wrong value for sensors

So VDDA is very sensitive to noise so must be clean and need to separate out the power supply from VDD

WHY FERRITE BEAD IS CONNECTED ONLY WITH VDDA SECTION

As the ferrite bead will stop noise to enter into digital section and will provide clean path

Digital circuit needs fast switching and instant current if ferrite bead is connected with digital

circuit then it will resist high current frequency and will slow down the current delivery therefore

don't connect ferrite bead with it

PIN HEADER FOR SWD, USART, I2C

For debugging perspective there is swd pin header connection

Usart, i2c are for external connection of sensors

PCB LAYOUT

2 layer pcb

The track width for power rail is 0.5mm which is capable of carrying current up to 2A
And other signal track width is 0.3mm

Front copper layer and bottom copper layer is apply with gnd solid plane

GND vias connection

Every gnd pad is connected with gnd vias so that the return current path loop remain short and tight